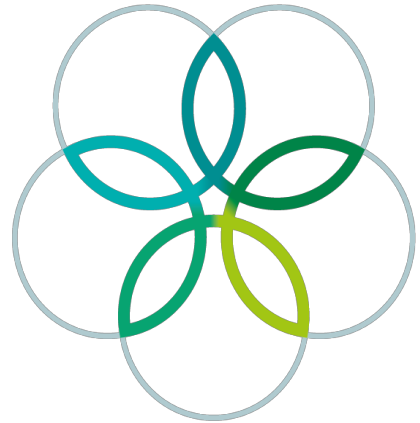
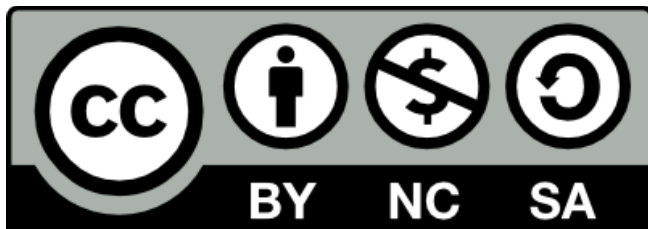


INTERNATIONAL
BIOLOGY
OLYMPIAD e. V.

IBO



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I INTERNATIONAL BIOLOGY OLYMPIAD

1. A relict is:

- a) a rapidly spreading organism that is not indigenous to the given area;
- b) the term for an organism causing dangerous disease;
- c) a genetically unstable hybrid;
- d) an organism which in the given area is a remainder from earlier times, when it was widespread.

2. As a result of intensive nitrogen use of fertilizers this one has spread to a number of biotopes which are not typical of it:

- a) coltsfoot (*Tussilago far far a*);
- b) viper's bugloss (*Echium vulgare*);
- c) great nettle (*Urtica dioica*);
- d) dandelion (*Taraxacum officinale*).

3. The highest gross primary production is typical of the ecosystem:

- a) beech wood; b) meadow; c) mountain pine forest; d) oak grove.

4. Water organisms characterized by striking active movement are known as:

- a) plankton; b) benthos; c) nekton; d) pleuston.

5. Indicate which ecological group is referred to the plants which have a thin, fragile stalk with reduced mechanical tissue, inside the stem are large intercellular spaces, the leaves are fleshy with a thin cuticle...

- a) mesophyte; b) hygrophyte; c) xerophyte; d) thermophyte.

6. The marking of territory is particularly significant for:

- a) finding a partner; b) genetic adaptation;
- c) a successful struggle for survival; d) competition within a species.

7. Putrefactive bacteria belong to:

- a) producers; b) consumers of the 1-st order;
- c) consumers of the 2-nd order; d) destroyers (reducers).

8. Acid rains have negative influence on nature:

- a) influence the quality of solar radiation;
- b) increase the movement of Ca^{2+} and certain metals in the ecosystem;
- c) increase the amount of H_2O_2 in the atmosphere;
- d) cause smog.

9. The eutrofication of water is caused by high content of:

- a) potassium and CO_2 , in the water;
- b) nitrogen and phosphorous in the water;
- c) sodium and calcium in the water;
- d) saccharides in the water.

10. The dominant of the herbal layer of an oak and hornbeam wood may be:

- a) deadly nightshade (*Atropa belladonna*);
- b) rejuvenating woodland beech (*Fagus sylvatica*);
- c) purple orchis (*Orchis macidata*);
- d) starwort (*Stellaria holostea*).

11. A mountain spruce forest is, in comparison with a field of maize:

- a) more stable, does not require additional energy to survive;
- b) more stable, but requires considerable input of additional energy;
- c) less stable, requires great amount of additional energy to survive;
- d) less stable, but due to high capacity for self regulation does not need additional energy.

12. Solar energy is best used per surface unit by:

- a) a field of wheat;
- b) a meadow;
- c) a pine forest;
- d) a water meadow.

13. The silver birch (*Betula pendula*) belong to the trees which are:

- a) strongly light loving;
- b) shade loving;
- c) indifferent to light;
- d) light and shade loving.

14. Mushrooms (Fungi) have the following relationship to light:

- a) they require it unconditionally for growth;
- b) they do not require it for growth, but many types need a certain amount of light for spore production;
- c) they have no need whatever of light for existence;
- d) apart from exceptions they require it (light has a positive influence, for instance, on rate of growth and colouring of gills).

15. Light for the germination of seeds is:

- a) necessary only for some plant species;
- b) quite essential for all parasitic plants;
- c) an essential factor only for light loving plants;
- d) is not a limiting factor for germination in any species.

16. Relation of insufficient light and yellowing of leaves:

- a) no influence;
- b) slows it down, especially at higher temperatures;
- c) slows it down, especially at lower temperatures;
- d) speeds it up, especially at higher temperatures.

17. Algae and infusoria live together in a solution placed in the light. The infusoria consumes 0,10 mol of glucose in a week, the algae - 0,12 mol. The weekly production of glucose amounts to 0,25 mol. What is the net production of oxygen per week in this medium?

- a) 0,03 mol;
- b) 0,60 mol;
- c) 1,32 mol;
- d) 0,18 mol.

18. "Synanthropic" is the term used for:

- a) enriching of local flora and fauna with species from other geographical territories;
- b) domestic (domesticised) animal;
- c) organisms transferred by man from original territories;
- d) animals and plants living in close relationship with man.

19. Phototaxis is:

- a) the ecological link of a certain species to a shaded or unshaded site;
- b) the amount of light in a certain unit of time required for the development of a given species;
- c) orientation (e.g. of insects) evoked by light;
- d) relation between growth of insect larvae and light conditions.

20. What is an edaphone?

- a) instrument for repelling insects (e.g. mosquitoes) with sound waves;
- b) community of all organisms living in the soil;
- c) part of the sound apparatus of a grasshopper;
- d) part of the sound apparatus of a mosquito.

21. A monophagous is:

- a) a cell capable of swallowing up foreign particles (e.g. microorganisms);
- b) an animal which takes in food only during a single stage of development;
- c) an animal with the highest degree of specialization as regards type of food;
- d) a virus attacking only narrowly specialized eucaryotic cell.

22. A predator is:

- a) a beast of prey living wild which kills and eats other animals;
- b) an animal living on or in the body of another animal and feeding on it;
- c) an animal which on or in its body carries the germs or development stages of other animals;
- d) a species of animal whose existence signals the later existence of another species.

23. A pheromone is:

- a) a chemical substance, containing mainly iron compounds, from which the body covering of insect is made;
- b) a hormone enabling relations between individuals of one species of insects;
- c) a chemical dye based on iron which gives rise to the metallic sheen of some groups of insects;
- d) a hormone enabling relations among individuals of various species of insects.

24. The term "mimicry" means:

- a) the microscopic feeler hairs on an insect's body;
- b) the special protective similarity of some species of insects to others;
- c) the state into which some species of insects fall under the influence of external stimuli signalling danger;
- d) the behaviour of the females of some groups of insects, intended to attract the male.

25. By the concept of "population" we mean:

- a) a set of individuals of one species including their developmental stages on a certain territory;
- b) a set of individuals of one species on a certain territory without stages of development;
- c) set of individuals of all species in a common environment;
- d) set of individuals on a certain territory which are not genetically related.

26. The genetic balance of a population in the sequence of generations is expressed by the Hardy-Weinberg Law, which is expressed mathematically as:

- a) $H = 2pq$;
- b) $p^2 + 2pq + q^2 = 1$;
- c) $(p+q) - (p - q)$;
- d) $(p+q) \times (p - q) = p^2 - q^2$.

27. The basic unit of an eucaryotic chromosome consisting of DNA and protein is:

- a) a nucleotide;
- b) a nucleoside;
- c) a nucleosome;
- d) a nucleoid.

28. All the given factors are mutagenous with the exception of:

- a) gamma-radiation;
- b) UV-radiation;
- c) acridine dye;
- d) acetic acid.

29. A caryotype is:

- a) the crossing of chromosomes during meiosis;
- b) the marking of sex chromosomes;
- c) characteristics of type of cell core (procaryotic or eucaryotic type);

30. In a monohybrid with complete dominance the phenotype splitting proportion in F_2 , is:

- a) 3:1;
- b) 1:2:1;
- c) 9:3:3:1;
- d) 1:1.

31. With regard to sex chromosomes a normal woman is equipped with:

- a) XY;
- b) XX;
- c) YY;
- d) XO.

32. A new pink flowering plant with genome Cc was planted on an island. It is an annual, self pollinating plant that has four seeds a year. If in its progeny there appears a plant with white flowers, then this plant is:

- a) polyploid;
- b) homozygotically recessive;
- c) a hybrid with incomplete dominance;
- d) a mutant.

33. In a case where one parent has blood group 0 and the other - AB, the children may belong to the group:

34. In the interbreeding of individuals with genotypes AABb and aaBb the following genotype cannot occur in further generations:

- a) AABb; b) AaBb; c) AaBB; d) Aabb.

35. The biogenetic law (onthogenesis is an abbreviation of phylogenesis) was formulated by:

- a) E. Haeckel; b) G. Cuvier; c) J.B. Lamarck; d) Ch. Darwin.

36. Among the oil soluble vitamin is:

- a) vitamin K; b) pyridoxine; c) thiamin; d) vitamin C.

37. Which tree releases allelopathic substances to the environment?

- a) beech (*Fagus sylvatica*); b) sallow (*Salix caprea*);
c) bastard acacia (*Robinia pseudacacia*); d) small-leaved lime (*Tilia cordata*).

38. During which process most energy quantity is released?

- a) photolysis; b) glycolysis; c) Krebs cycle (the cycle of citric acid);
d) final oxidation in the respiration chain.

39. Some substances enter a photosynthesising cell and some leave the cell:

cell:

- a) glucose and CO₂ — water and O₂;
b) glucose and O₂ — water and CO₂;
c) water and O₂ — glucose and CO₂;
d) water and CO₂ - glucose and O₂.

40. Which of the given actions are linked with the primary processes of photosynthesis (with the light phase):

The codes of the answers: 1. photolysis; 2. Calvin cycle;

3. elimination O₂; 4. absorption CO₂.

- a) 1, 2; b) 1, 3; c) 2, 3; d) 2, 4.

41. The energetic effect of anaerobic glycolysis are 2 molecules of:

- a) lactic acid; b) pyruvic acid; c) ATP; d) ethanol.

42. In the photosynthesis of C₃-plants the acceptor of CO₂ is:

- a) 3-phosphoglycerol aldehyde; b) ribulose-1,5-diphosphate;
c) phosphoenolpyruvic acid; d) ferredoxin

43. Photorespiration is a process typical for:

- a) all plants without exception in roughly the same intensity;
b) mainly C₄-plants;
c) mainly C₃-plants;
d) this process is not typical for plants.

44. The following has a marked capacity for vegetative reproduction:

- a) orache (*Atriplex*); b) plantain (*Plantago*);
c) coltsfoot (*Tussilago*); d) chicory (*Cichorium*).

45. Gametophyte is in the life of plants a generation which:

- a) creates gametes through the process known as meiosis;
b) arises from gametes;
c) arises as a result of the combining of gametes;
d) is created by cells with a haploid number of chromosomes in the nucleus.

46. Not among the flowering plants is the:

- a) hazel (*Corylus*); b) poplar (*Populus*);
- c) larch (*Larix*); d) walnut (*Juglans*).

47. An angular stem, symmetrical flowers in tight inflorescences, two longer and two shorter stamens characterise the family of: 55

- a) mustard (*Brassicaceae*); b) poppies (*Papaveraceae*); th
- c) dead-nettles (*Lamiaceae*); d) figworts (*Scrophulariaceae*).

48. Willows (*Salix*) are pollinated by... and their seeds are spreaded by..

- a) the wind - the wind; b) the wind - animals;
- c) insects - water or animals; d) insects - the wind.
- d) characteristics giving number, shape and construction of all chromosomes of an individual.

49. Mycelium is not formed by:

- a) *Mucor mucedo*; b) *Claviceps purpurea*;
- c) *Puccinia graminis*; d) *Saccharomyces cerevisiae*.

50. A common trait of *Cyanophyta* and *Rhodophyta* is:

- a) more or less the same cell construction;
- b) more or less the same colouring;
- c) production of starch during photosynthesis;
- d) in reproduction similar single cell units are formed.

51. Apart from the single cotyledon monocotyledonous plants can be further characterised by the following traits:

- a) cambium, replacement roots, perianth;
- b) cambium, main root and side roots;
- c) scattered vascular bundles, main root and side roots, perianth;
- d) scattered vascular bundles, replacement roots.

52. The flower of buttercup (*Ranunculus*) is characterised by:

- a) arrangement of K and C in ring, A and G - in spiral, free pistils;
- b) arrangement of P in ring, A and G - in spiral, free pistils;
- c) arrangement of all parts in ring, free pistils;
- d) arrangement of all parts in spiral, pistils grown into the receptacle.

53. In plants with enclosed seeds the seeds originate from:

- a) the ovary and sepals; b) the ovary;
- c) the ovule; d) the fertilized egg cell.

54. Diatoms (*Bacillariophyta*) are single cell organisms:

- a) feeding mainly heterotrophically and living in colonies;
- b) moving with the aid of flagella or pseudopods;
- c) forming a double cell wall of CaCO_3 ;
- d) forming a cell wall of SiO_2 .

55. With the transition of plants from a watery environment to the dry land there occurred in phylogenesis mainly the improvement of:

- a) mechanical tissues and the methods of reproduction;
- b) assimilation tissues and conductive tissues;
- c) covering tissues and assimilation tissues;
- d) methods of reproduction, conductive and mechanical tissues.

65. The respiratory muscles of a man are:

- a) the internal crosswise stomach muscle and diaphragm;
- b) the external and internal intercostal muscles and direct stomach muscle;
- c) the external and internal intercostal muscles and diaphragm;
- d) the large and small chest muscle and diaphragm.

- 66. When at rest the surface of the membrane of a nerve cell is:**
 a) electropositive; b) electronegative; c) electroneutral; d) without electric charge.
- 67. The composition of lymph (sap) is similar to the composition of blood plasma. Select the correct statement:**
 a) proteins are lacking in lymph;
 b) there are more proteins in honph than in plasma;
 c) there are less proteins in lymph than in plasma;
 d) the difference is in the quality of Na⁺ ions.
- 68. The semilunar valves separate:**
 a) the chambers of the heart from the arteries; b) the left ventricle from the right atrium;
 c) the right ventricle from the left atrium; d) hollow veins from the right atrium.
- 69. Progesteron is eliminated by:**
 a) the Graafian follicle; b) the corpus luteum (Yellow body);
 c) the maturing egg; d) the ovulating egg.
- 70. The biosynthesis of proteins takes place in all cells of the human organism with the exception of:**
 a) the cells of the pancreatic gland;
 b) the cells of the mucous membrane of the intestine;
 c) mature red corpuscles (erythrocytes);
 d) white blood corpuscles (lymphocytes).
- 71. The veins (trachea) of the altered root of mistletoe (*Viscum*) grow through a pine (*Pinus*) branch, for instance, into the:**
 a) pithy part of the vascular bundle (phloem);
 b) woody part of the vascular bundle (xylem);
 c) pith of the branch;
 d) immediate surroundings of individual vascular bundles, but do not penetrate them.
- 72. Sexual reproduction in Infusoria with gene recombination takes place by:**
 a) encystation; b) strobilation; c) conjugation; d) longitudinal splitting.
- 73. The mesoderm appears first in phylogenesis in the:**
 a) *Scyphozoa*; b) *Turbellaria*; c) *Tracheata*; d) *Annelida*.
- 74. The symbiosis between termite (*Isoptera*) and the microorganisms, living in their intestines and feeding on cellulose is an example of:**
 a) competition; b) predation; c) mutualism; d) commensalism.
- 75. Various tissues and organs are formed from the ecto-, meso- and endoderm. Which combination is correct?**
- | | <i>ectoderm</i> | <i>mesoderm</i> | <i>endoderm</i> |
|----|------------------------|-----------------|-----------------|
| a) | brain and spinal fluid | blood | lungs |
| b) | brain and spinal fluid | large intestine | lungs |
| c) | skin | bone | kidneys |
| d) | epidermis | liver | heart |
- 76. The human organism does not synthesise and must therefore require from food:**
 a) purines; b) pyrimidines; c) fructose; d) aromatic amino acids.
- 77. Caterpillars of the cabbage white (*Pieris brassicae*) form a chrysalis:**
 a) underground near a nutrition's plant;
 b) on various objects above ground (trees, posts, walls);

- c) on a food plant;
- d) do not make a chrysalis because it is an insect with imperfect metamorphosis.

78. Colorado beetle (*Leptinotarsa decemlineata*) was brought to Europe from North America in:

- a) 1925; b) 1938; c) 1947; d) 1914.

79. Cricket (*Gryllidae*) chirp:

- a) by rubbing their front wings together;
- b) by rubbing the front wing against the back leg;
- c) by means of a special organ on the lower side of the thorax;
- d) by pushing collected air out through the stigmata.

80. Rotifers (*Rotatoria*), like all roundworms (*Nemathelminthes*), do not have a developed system.

- a) digestive; b) eliminative; c) respiratory; d) nervous.

81. Halteres are:

- a) moisture loving ribbon like shapes of the spores of horsetails;
- b) part of the mandibles of some beetles;
- c) the reduced wings, for example, of the mosquitos;
- d) the sensitive cilia on the rear part of the body of homoptera.

82. Which type of parasitic worms has a stage of development with life in fresh C water?

- a) roundworm (*Ascaris lumbricoides*);
- b) nematode worm (*Trichinella spiralis*);
- c) beef tapeworm (*Taeniarhynchus saginatus*);
- d) liver rot (*Fasciola hepatica*).

83. Parthenogenesis can occur in all the following organisms with the exception of:

- a) rotifers (*Rotatoria*); b) insects (*Insecta*);
- c) crustaceous (*Crustacea*); d) spiders (*Arachnida*).

84. What is the largest number of elements of which an insect's leg may be made up?

- a) 5; b) 6; c) 9; d) 15.

85. *Planorbis* breathe:

- a) through gills hidden by part of the shell so that they are actually within the body;
- b) through two external gills;
- c) through a lung sac formed from the wall of the shell hollow;
- d) through a single gill (*ctenidium*) near the anal opening.

86. Chorda (*chorda dorsalis*) is preserved in mature age by:

- a) sea squirts (*Ascidiae*); b) sturgeon (*Acipenser*);
- c) lamprey (*Petromyzones*); d) dipnoan fish (*Protopterus*).

87. The most primitive placental mammals are:

- a) insect-eating mammals (*Insectivora*); b) marsupials (*Marsupialia*);
- c) rodents (*Rodentia*); d) protothere (*Monotremata*).

88. In herbivores the food returns to the mouth for cudling from:

- a) the crop (*ingluvies*); b) the honeycomb (*reticulum*);
- c) the second stomach (*rumen*); d) the psalterium (*omasum*).

89. European species of amphibians live on a diet which is:

- a) mainly animal; b) mainly vegetable;
- c) for the young - vegetable, for adult - animal;

d) for young frogs - mixed, for others - animal.

90. The brown frog (*Rana temporaria*) has teeth:

- a) on the upper and lower jaws;
- b) on the upper jaw and roof of mouth;
- c) on the lower jaw;
- d) has no teeth at all.

91. The cloaca is:

- a) the sexual organ of fish, amphibians and some reptiles;
- b) a spiral surface in the intestines of sharks and some fish;
- c) a common opening for the digestive, eliminator)' and sex organs;
- d) in water birds a gland serving for the oiling of feathers.

92. European species of lizard can threaten the health of man:

- a) by biting with teeth and poison gland;
- b) as an intermediary host to some intestinal flatworms (*Plathelminthes*);
- c) by biting and infecting wound with poisonous secretions from the mouth;
- d) they are not at all dangerous.

93. Zoochoria is:

- a) the broadcasting of seeds or fruits by activity of animals;
- b) an infectious disease carried by animal fur;
- c) the science of the spreading of animals on Earth;
- d) use of animals for "biological warfare" with other animal species.

94. In European amphibians the fertilization of eggs takes place:

- a) in the water outside the female's body;
- b) in the female's body;
- c) in frogs - outside the female's body, in salamanders and newts - inside the body
- d) in frogs and newts - in the water, in salamanders - in the female's body.

95. The nest of the great tit (*Parus major*) is usually situated:

- a) in the fork of a branch of a thick bush;
- b) mostly under a bush on the ground in grass;
- c) is usually woven of grass stalks and hung on a branch 3 m above ground;
- d) in a hollow tree.

96. From the following mammals one does not hibernate:

- a) great dormouse (*Glis glis*);
- b) european hedgehog (*Erinaceus europaeus*);
- c) common mole (*Talpa europaea*);
- d) long-eared bat (*Plecotus auritus*).

97. Which bird of prey lives mainly on smaller birds?

- a) buzzard (*Buteo buteo*);
- b) marsh harrier (*Circus aeruginosus*);
- c) kestrel (*Falco tinnunculus*);
- d) sparrow hawk (*Accipiter nisus*).

98. Articulates (*Arthropoda*) are linked in development to:

- a) *Mollusca*;
- b) *Nemathelminthes*;
- c) *Annelida*;
- d) *Echinodermata*.

99. One of the following vertebrate animals does not have ribs:

- a) newt (*Thyturus*);
- b) bat (*Myotis*);
- c) sparrow (*Passer*);
- d) mole (*Talpa*).

100. L. Pasteur discovered:

- a) vaccination against smallpox;
- b) blood-coagulation factors;
- c) the treatment of rabies;
- d) bacteriophages.

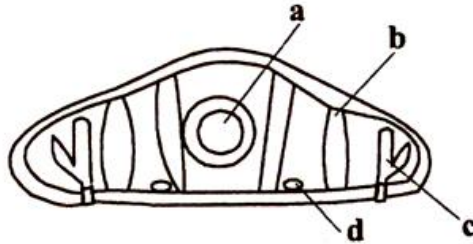
***Following Information deals with the questions №101-104.

Body construction of *Turbellaria* (101), *Annelida* (102), *Crustacea* (103), *Insecta* (104) is represented on the figures. Numerate on the schemes:

The codes of the answers:

- | | |
|------------------------|----------------------|
| 1. nervous system; | 3. vascular system; |
| 2. alimentary channel; | 4. secretory system. |

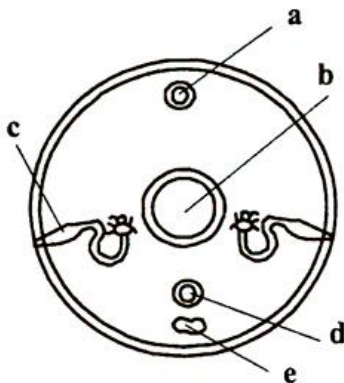
101.



Answers:

a) ...; b) ...; c) ...; d) ...; e)

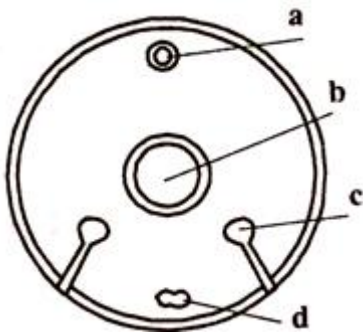
102.



Answers:

a) a) ...; b) ...; c) ...; d) ...; e)

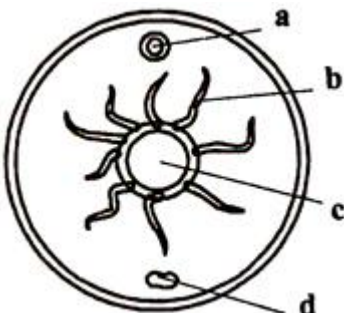
103.



Answers:

a) a) ...; b) ...; c) ...; d) ...; e)

104.



Answers:

a) a) ...; b) ...; c) ...; d) ...; e)

*****Following information deals with the questions №105-108.**

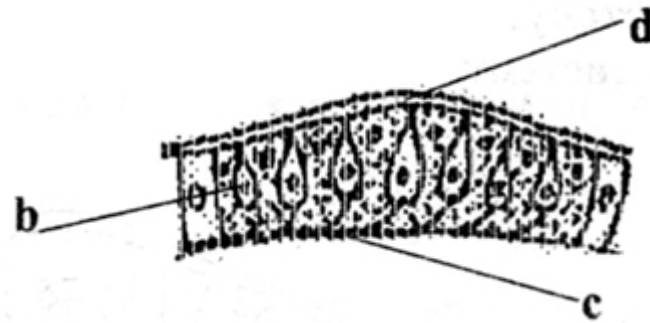
Some types of eyes are represented on the figures: *Turbellaria* (105), *Insecta* (106), *Cephalopoda* (107) and *Mammalia* (108). Numerate on the schemes:

The codes of the answers:

- | | |
|--------------------------|------------------------------|
| 1. nervous system; | 3. layer of pigmental cells; |
| 2. photosensitive cells; | 4. lens. |

105.

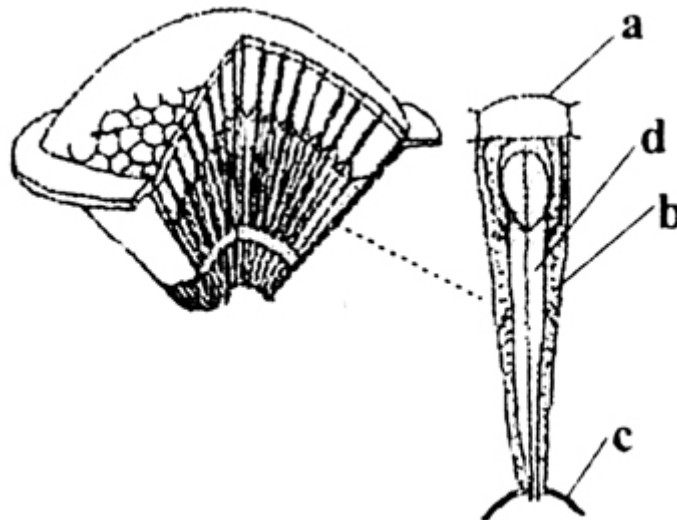
Answers:



a)...; b) .. c).. d)

106.

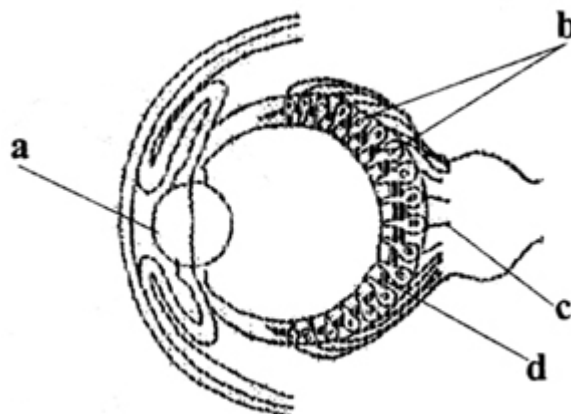
Answers:



a)...; b) .. c).. d) ...

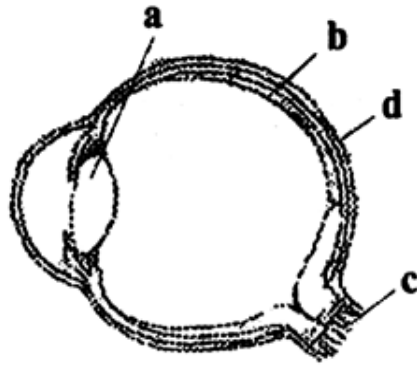
107.

Answers:



a)...; b) .. c).. d)

108.



Answers:

a)...; b) .. c).. d)

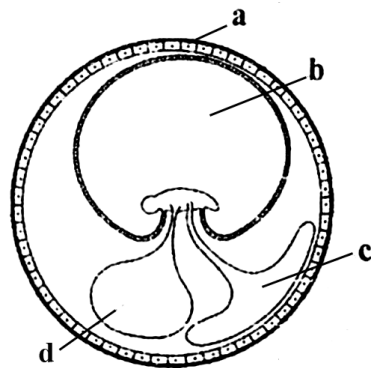
*****Following information deals with the questions №109-110.**

The are schemes of embryogeny of bird (109) and mammal (110) are represented on the figures. Numerate on the schemes:

The codes of the answers:

- | | |
|---------------|----------------------|
| 1. yolk sac; | 4. chorion; |
| 2. amnion; | 5. navel string; |
| 3. allantois; | 6. chorionic villus. |

109.



Answers:

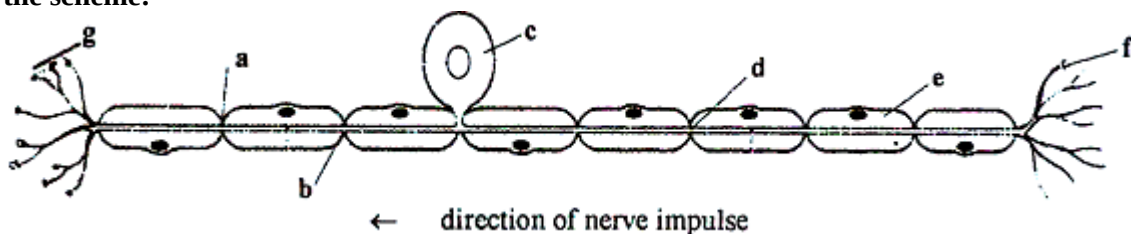
a)...; b) c) d)



110. Answers:

a)...; b) c) d) e) f)

111. Scheme of construction of sensitive neuron represented on the figure. Numerate on the scheme:



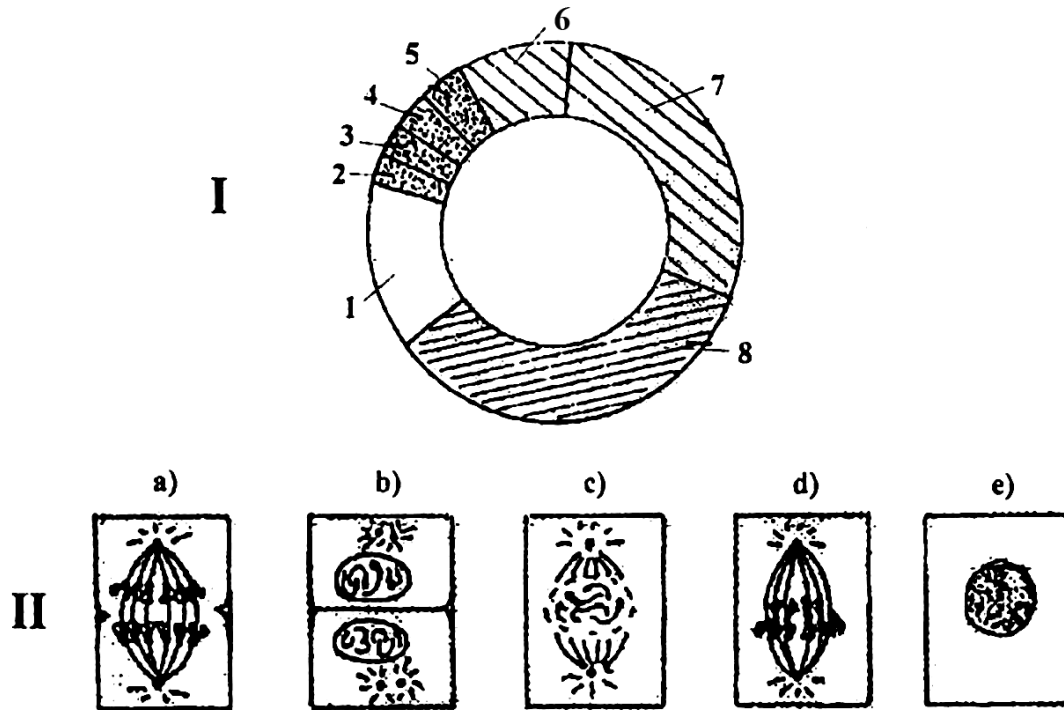
The codes of the answers:

1. cell body;
2. axon;
3. dendrite;
4. Schwann cell;
5. node of Ranvier;
6. presynaptic membrane;
7. postsynaptic membrane.

Answers:

a) b) c) d) e) f) g)

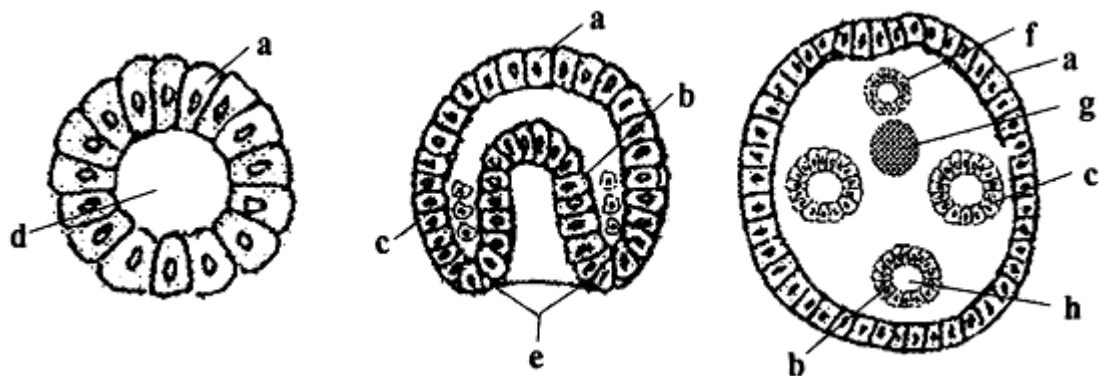
112. Stages of the cell's cycle are marked by numbers. Find the according between the numerals on the figure I and letters, which denote the stages of cell's cycle, on figure II.



Answers:

a) b) c) d) e)

113. Numerate on the figure infancies of lancelet:



The codes of the answers:

1. blastocoel;
2. blastopore;
3. archenteron;
4. ectoderm;
5. endoderm;
6. mesoderm;
7. building of nervous system;
8. chorda.

Answers:

a) ...; b) c) d) e) f) g) h)

114. Cycle of the individual development of moss is represented on the figure. Numerate on the scheme:

The codes of the answers:

1. cells with $2n$ set chromosomes (sporophyte);
2. cells with n set chromosomes (gametophyte);
3. archegonium;
4. anteridium.

Answers:

a)...; b) c) d) e) f) g)

115. Cycle of the individual development of lichen is represented on the figure. Numerate on the scheme:

The codes of the answers:

1. cells with $2n$ set chromosomes (sporophyte);
2. cells with n set chromosomes (gametophyte);
3. strobilus;
4. microspores;
5. megaspores;
6. archegonium;
7. anteridium.

Answers:

a)...; b) c) d) e) f)...; g) h)...; i) j)

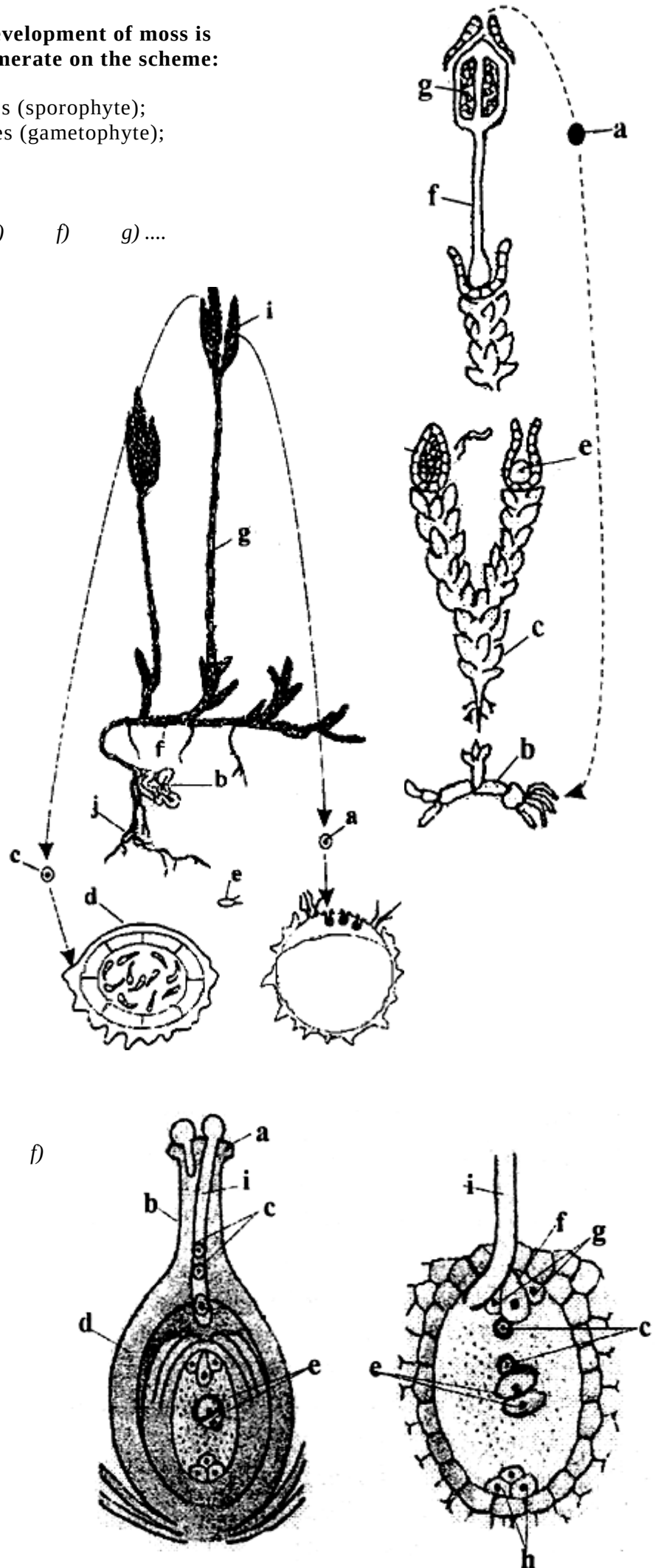
116. Cross-cut of pistil is represented on the figure. Name its parts with the help of the codes:

The codes of the answers:

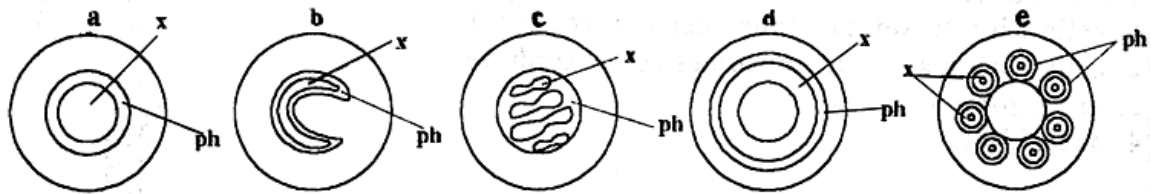
1. egg;
2. sperm;
3. polar nucleus;
4. stigma;
5. ovary;
6. synergids;
7. antipodal cells;
8. style;
9. pollen tube.

Answers:

a)...; b) c) d) e) f) g) h) i)



117. There are schemes of situation of the conductive textures in the cross-cuts of stalk of extinct rhynia (1), of club-moss (2), of horsetail (3), of fern (4) and pine (5). Number on the schemes (ph - phloem, x - xylem).

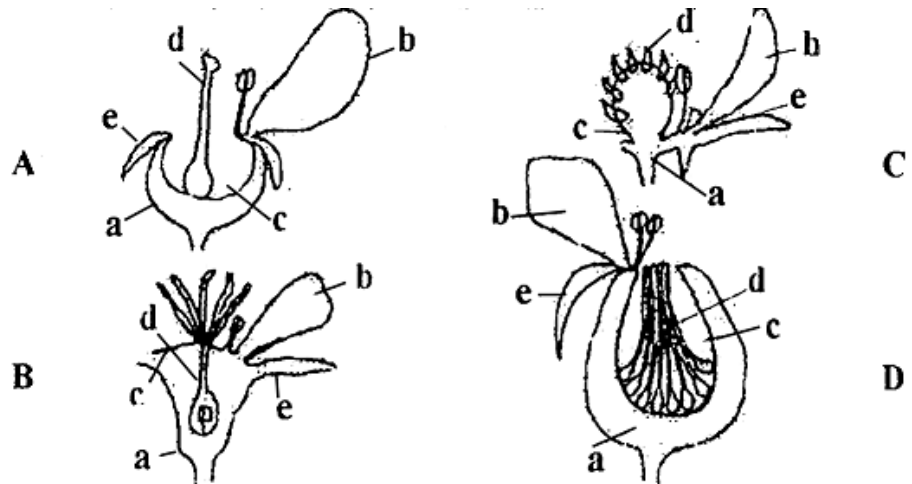


Answers: a) b) c) d) e)

118. On the figures fore-and-aft crosses of flowers of rosace (*Rosaceae*). Also compliment the numbers of the figures.

The codes of the answers:

1. nectary;
2. corolla;
3. cup;
4. pistil;
5. receptacle;
6. strawberry;
7. pear;
8. rose;
9. plum.



Answers: a) ...; b) c) d) e) A) ...; B) C) D)

119. Outgoing from the ecological regularity, complement the Figure into squares with the denominations of such plant's species, which can constitute, as a result, fragment of defoliated forest with natural species composition. Order of designation of species in a separate several horizons are not important.

The codes of the answers:

1. durmast oak (*Quercus petraea*);
2. small-leaved lime (*Tilia cordata*);
3. European hazel (*Corylus avellana*);
4. weymouth pine (*Pinus strobus*);
5. golden osier (*Salix alba*);
6. European hornbeam (*Carpinus betulus*);
7. loranth (*Loranthus europaeus*);
8. juniper common (*Juniperus communis*);
9. frail (*Phragmites australis*);
10. daphne february (*Daphne mezereum*);
11. annual meadow grass (*Poa annua*);
12. fescue (*Festuca rubra*);
13. nightshade black (*Solanum nigrum*);
14. chickling (*Lathyrus vernus*);
15. mustard dish (*Thlaspi arvense*);
16. wood bluegrass (*Poa nemoralis*);
17. *Symphoricarpos alba*;
18. *Nardus strica*.

Answers: a) b) c) d)

